

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	15 June 2026
Prepared By	Sowmya Kella
Project Name	ShopEZ: E-Commerce Application
Maximum Marks	4 Marks

Technical Architecture:

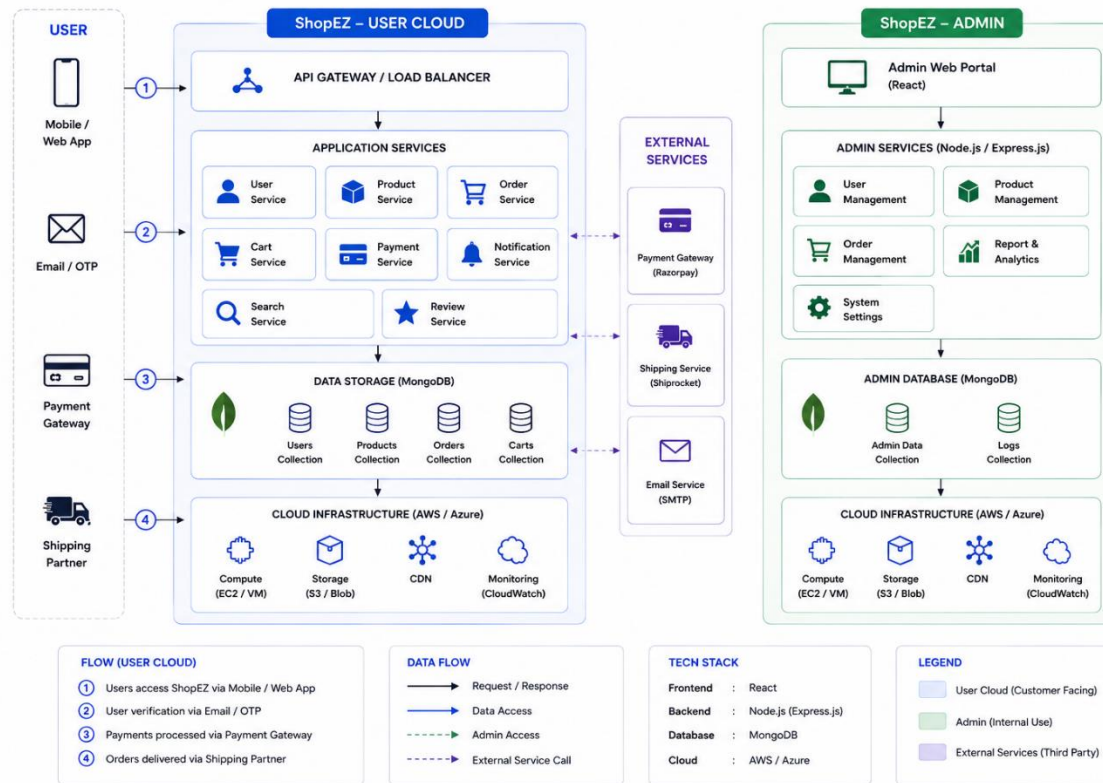


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based interface for users to browse products, manage cart, and place orders	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	User authentication, product management, cart and order processing	Java / Python
3.	Application Logic-2	Product CRUD operations (Add, Update, Delete, View Products)	IBM Watson STT service
4.	Application Logic-3	Admin dashboard functionalities and order management	IBM Watson Assistant
5.	Database	Stores user accounts, products, cart data and orders	MySQL, NoSQL, etc.
6.	Cloud Database	Cloud-hosted MongoDB database	IBM DB2, IBM Cloudant etc.
7.	File Storage	Stores product images	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	No external API used	N/A
9.	External API-2	No external API used	N/A
10.	Machine Learning Model	Not applicable for this application	N/A
11.	Infrastructure (Server / Cloud)	Backend hosted on Node.js server and database hosted on cloud	Local Server, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for frontend and backend development	React.js, Node.js, Express.js
2.	Security Implementations	User authentication, password encryption, protected routes	JWT Authentication, bcrypt
3.	Scalable Architecture	Three-tier architecture separating frontend, backend and database layers	React.js, Express.js, MongoDB
4.	Availability	Application remains accessible through cloud-hosted server	MongoDB Atlas, Node.js Server database and dedicated backend
5.	Performance	Fast product retrieval and API communication using efficient database queries and REST APIs	MongoDB, Express.js REST APIs